

Deep Learning Based Sensor Fusion for Endpoint Prediction and Control of Transhumeral Prostheses

Introduction

- Upper limbs have over 30 DOFs (arm+hand) in total [1].
- Amputees are **unable** to coordinate multi-joint prostheses.
- Previous work has shown correlation between gaze, sEMG and hand motion (see Krausz et al.[2]).
- We propose a **multi-modal sensor fusion model** as an endpoint predictor for a prosthetic arm controller.

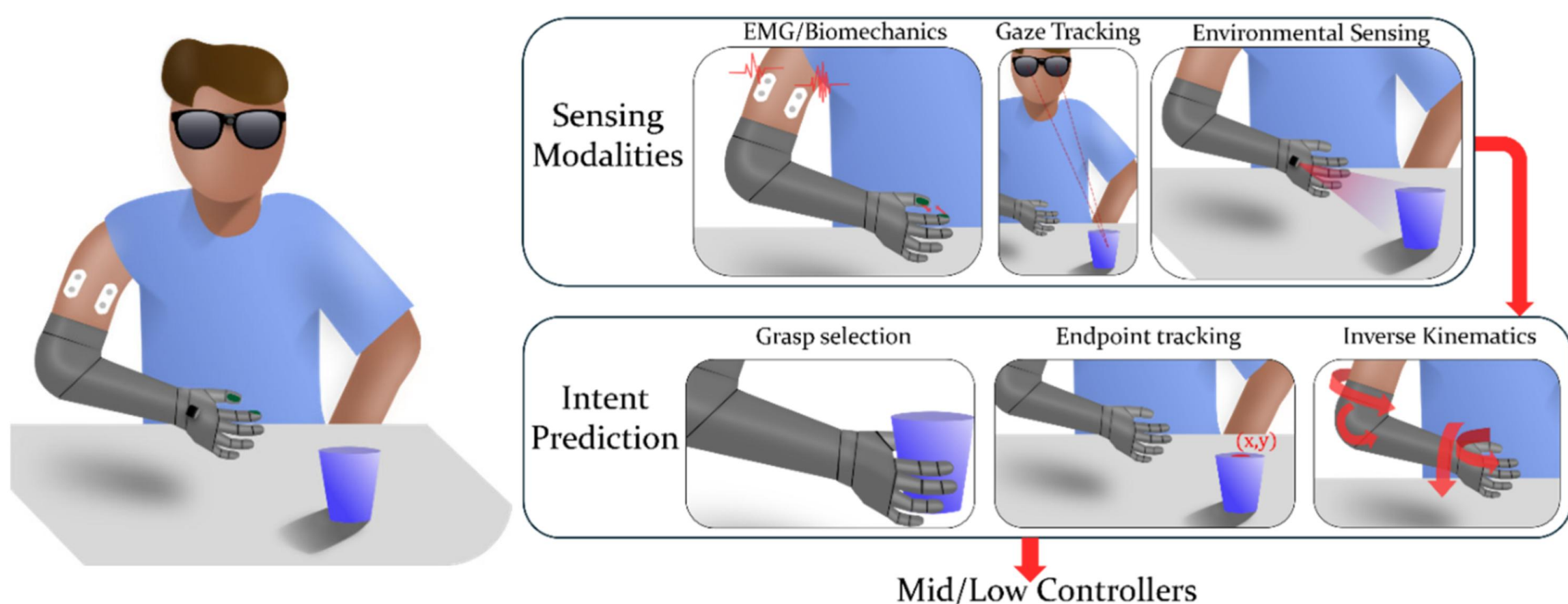


Fig 1. Multi-modal semi-autonomous prosthesis control concept from MOST BIONAC project

Methods

- Gathered **healthy pilot data**.
- Pre-processed the data and computed upper-arm **elevation angles** according to Soechting & Flanders [3].
- Built a **hybrid CNN & GRU PyTorch** model:
 - Inputs:**
 - L&R eye position, optical axis, head IMU**
 - elevation angles**
 - upper-arm and shoulder sEMG**
 - Outputs: 3D coordinates of palm, L&R sides of wrist.**
- Compared predicted coordinates to MoCap ground truth.

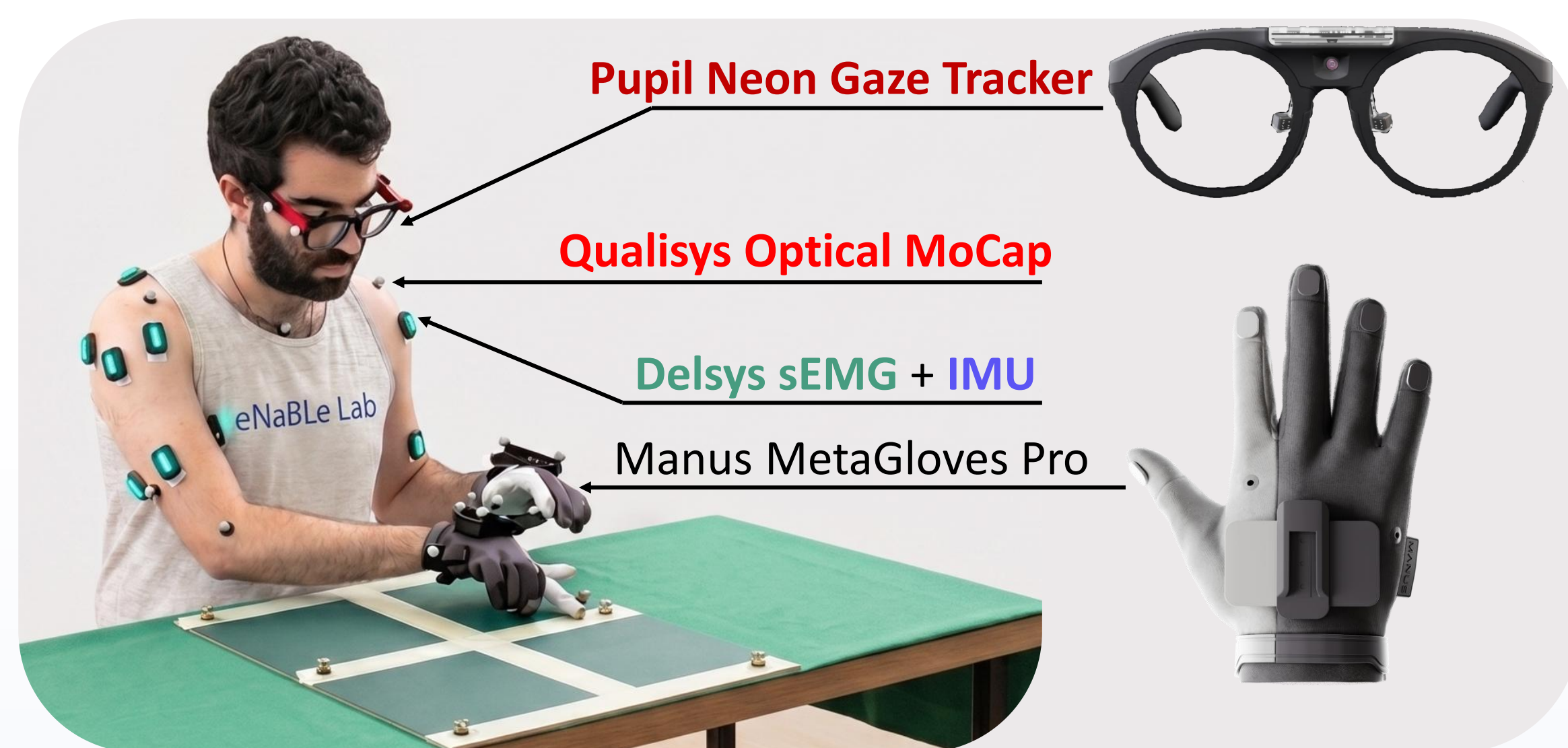


Fig 4. Illustration of the experimental setup and sensors

Future Work

- Acquire full dataset combining **MoCap, IMU, sEMG, and gaze-tracking** data recorded during simulated ADLs.
- Identify relevant **synergies** between sensor modalities.
- Create an efficient **real-time endpoint predictor and controller**.
- Control a robotic hand and FRANKA arm to demonstrate real-time **prediction of user intent**.
- Test controller with a transhumeral amputee user and custom prosthesis.

Preliminary Results

Average MAE When Using Different Sensors
All values in [mm], green=low MAE, red=high MAE

	Overall 3D	Palm			Wrist-In			Wrist-Out		
		Palm X	Palm Y	Palm Z	Wrist In X	Wrist In Y	Wrist In Z	Wrist Out X	Wrist Out Y	Wrist Out Z
Baseline (All Sensors)	65.53	45.46	31.63	22.76	38.07	30.61	25.62	40.14	33.78	23.78
sEMG Alone	99.78	75.49	45.54	27.73	62.67	41.60	27.72	67.21	45.40	28.20
Elevation Angles Alone	79.81	64.22	34.84	25.22	51.09	30.29	23.36	54.84	38.86	22.31
Optical Axes Alone	90.50	59.82	48.29	25.58	51.60	45.89	26.62	56.24	45.71	26.95
Eye Positions Alone	105.43	80.13	44.23	22.88	71.64	42.58	21.63	81.65	43.46	22.38
Head Accelerometer Alone	141.16	108.40	64.87	39.70	96.00	59.33	39.67	104.68	64.48	38.52
Head Gyro Alone	102.22	73.57	49.60	33.85	61.83	45.72	34.15	64.26	47.21	33.45
Control - No Sensors	108.03	79.99	50.84	41.76	66.08	45.82	39.94	67.70	50.84	39.30

Fig 2. Average MAE of the predicted endpoint under baseline (all sensor modalities), individual sensor, and no sensors conditions

- The baseline model performs surprisingly well, with an MAE of **6.5 cm** (Note: 50% percentile male hand width = 10.4 cm [4])
- Individual sensor modalities do not provide sufficient information for accurate prediction.
- The **eye positions**, head-mounted **gyroscope**, head-mounted **accelerometer**, and the **sEMG** sensor sites are not predictive on their own.

Ablation Study - Difference in Average MAE Relative to a Baseline (Using All Sensors)
All values in [mm], green=improvement, red=degradation

	Overall 3D	Palm			Wrist-In			Wrist-Out		
		Palm X	Palm Y	Palm Z	Wrist In X	Wrist In Y	Wrist In Z	Wrist Out X	Wrist Out Y	Wrist Out Z
Dropped Head Gyro	-0.33	-0.10	0.45	-0.71	-0.19	0.41	-2.06	1.12	-0.77	-0.73
Dropped Head Accelerometer	5.57	3.68	7.72	-3.26	3.85	7.53	-3.92	5.80	2.11	-2.28
Dropped Eye Positions	9.82	10.26	4.01	-0.05	8.34	4.55	-0.69	8.61	1.52	-0.43
Dropped Optical Axes	13.78	12.31	6.18	1.54	7.63	6.87	1.66	11.99	4.63	2.00
Dropped Elevation Angles	20.21	12.84	10.51	4.81	12.28	7.96	4.30	17.31	10.87	5.13
Dropped All sEMG	8.40	10.29	3.74	0.84	4.80	3.61	-1.75	6.33	4.08	-0.70
Control - Drop All Sensors	42.50	34.52	19.21	19.01	28.01	15.21	14.32	27.56	17.06	15.52

Fig 3. Ablation Results showing relative errors when dropping different sensor modalities VS baseline

- Elevation angles**, particularly *upper arm yaw* (η), are highly predictive of hand endpoint position.
- sEMG** from *Trapezius Descendens*, *Posterior Deltoid*, and *Biceps* contributed most amongst recorded muscles.
- Early analyses show that delaying certain sensors improves prediction accuracy.
 - Largest improvement is from delaying the head-mounted **gyro** 100[ms] - about 11.5% .

Proximal sensors and gaze eNaBLE prediction of hand motion

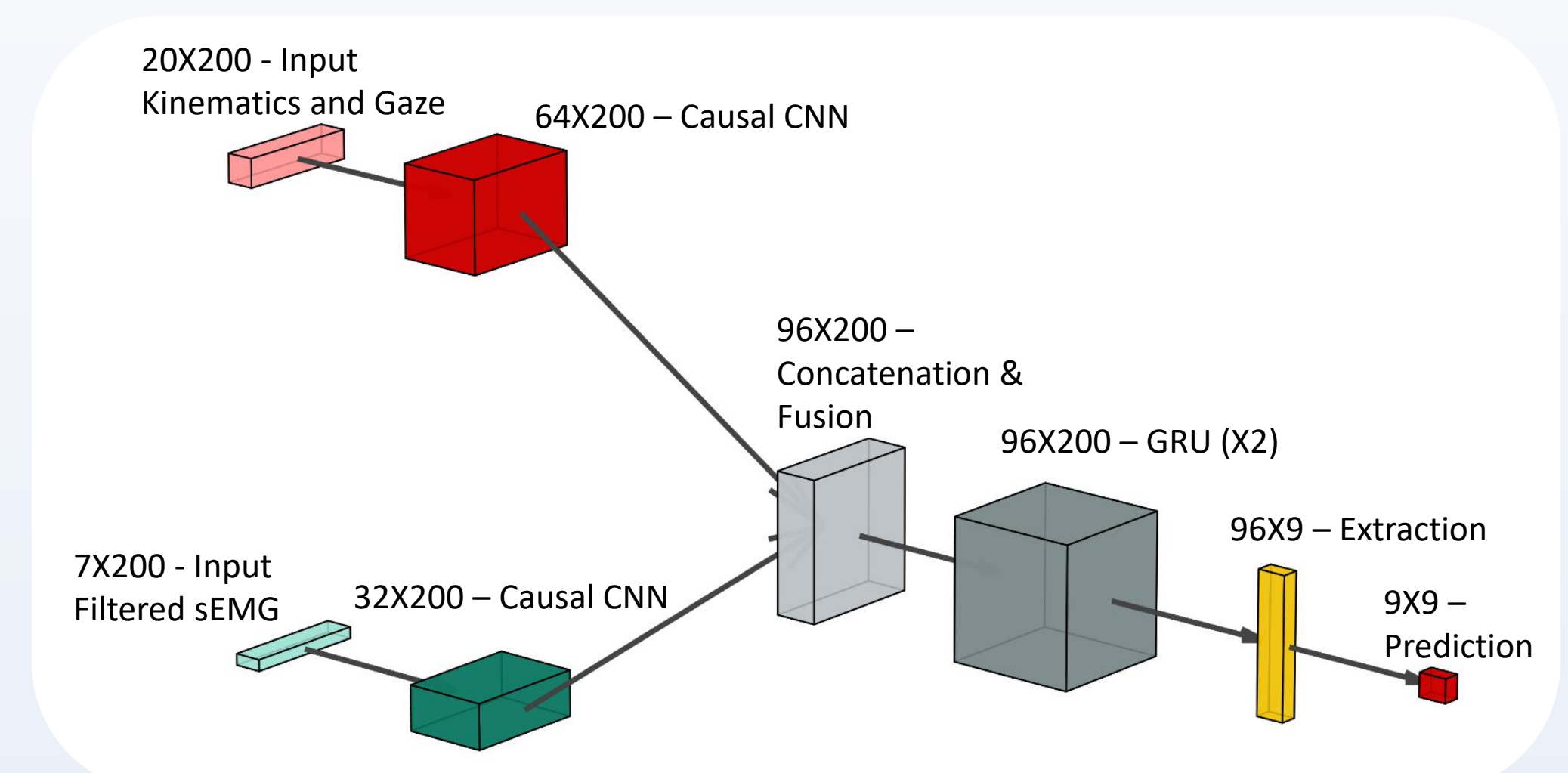


Fig 5. Illustration of our model's architecture